

📊 Hypothesis Testing & Statistical Inference

A Complete, Exam-Ready Study Guide

Hypothesis Testing · P-value · Z-test · Student's t-Distribution · T-test · Z vs T · Type I & II Errors · Bayes Theorem

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1. Hypothesis & Hypothesis Testing Mechanism

Inferential Statistics = using a **sample** to draw a **conclusion or inference** about a whole **population**. The core tool used to do this rigorously is called **Hypothesis Testing**.

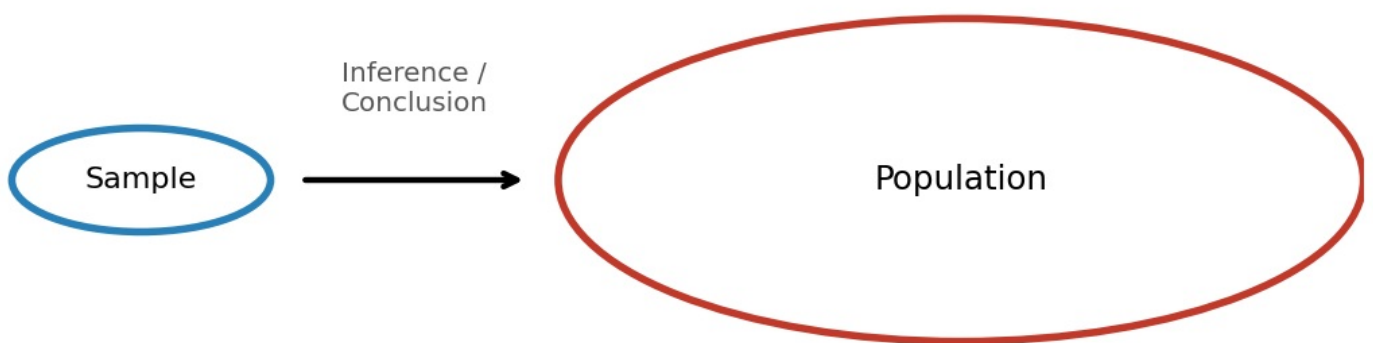


Fig 1.1 — We study a small sample and use hypothesis testing to infer something about the entire population.

Courtroom Analogy: Person → accused of Crime → goes to Court. The court assumes the person is "**not guilty**" until proven otherwise beyond reasonable doubt. This is exactly how hypothesis testing works!

1.1 The Hypothesis Testing Mechanism (4 Steps)

1. **Null Hypothesis (H_0):** The assumption you start with (the "status quo"). Courtroom: "the person is NOT guilty."
2. **Alternate Hypothesis (H_1):** The opposite of the null hypothesis — what you're trying to find evidence for. Courtroom: "the person IS guilty."
3. **Experiment → Statistical Analysis:** Collect data and compute test statistics {p-value, significance value} — direct proof (like DNA or fingerprint tests in a

courtroom).

4. **Decision:** Either **Accept the Null Hypothesis** (fail to reject H_0) OR **Reject the Null Hypothesis** (accept H_1).

Worked Example — College Pass Percentage

Colleges in District A state their average passed percentage of students is **85%**. A new college opens in the district, and a sample of 100 students shows a pass percentage of **90%**, with a standard deviation of 4%. Does this new college have a genuinely different pass percentage?

Null Hypothesis (H_0): $\mu = 85\%$ (no real difference — 90% is just sample variation)

Alternate Hypothesis (H_1): $\mu \neq 85\%$ (there IS a real difference)

(We'll finish solving this once we introduce the Z-test in Section 3.)

✱ Extra Example — Everyday Framing

A pharmaceutical company claims a new painkiller relieves headaches in an average of 20 minutes.

- **H_0 :** $\mu = 20$ minutes (the claim is true, drug behaves as advertised)
- **H_1 :** $\mu \neq 20$ minutes (the claim is false)

We would then test a sample of patients, measure their average relief time, and use statistics to decide whether to reject H_0 .

💡 We **never truly "prove" H_0** — we only ever say "fail to reject H_0 " (not enough evidence against it) or "reject H_0 " (strong evidence against it), just like a court says "not guilty" rather than "innocent."

2. P-value

Definition: The p-value is a number, calculated from a statistical test, that describes **how likely you are to have found a particular set of observations if the null hypothesis were true**. It's used to decide whether to reject H_0 .

💡 **Analogy:** Imagine tapping the spacebar 100 times and it lands in a certain region 20 times — that observed frequency corresponds to a probability, exactly like a p-value tells you "how likely is this outcome, assuming nothing unusual (H_0) is going on?"

2.1 Worked Example — Is the Coin Fair? (100 tosses)

1. **Null Hypothesis H_0 :** Coin is fair $\rightarrow \Pr(H) = 0.5, \Pr(T) = 0.5$
2. **Alternate Hypothesis H_1 :** Coin is NOT fair
3. **Experiment:** Toss the coin 100 times
4. **Significance Value:** $\alpha = 0.05 \rightarrow$ Confidence Interval $= 1 - 0.05 = 0.95$ (95%)
5. **Conclusion Rule:** If $p < \text{significance value } (\alpha) \rightarrow$ **Reject the Null Hypothesis**. Else $\rightarrow$ **Fail to reject the Null Hypothesis**.

Hypothesis Test — Is the Coin Fair? (100 tosses)

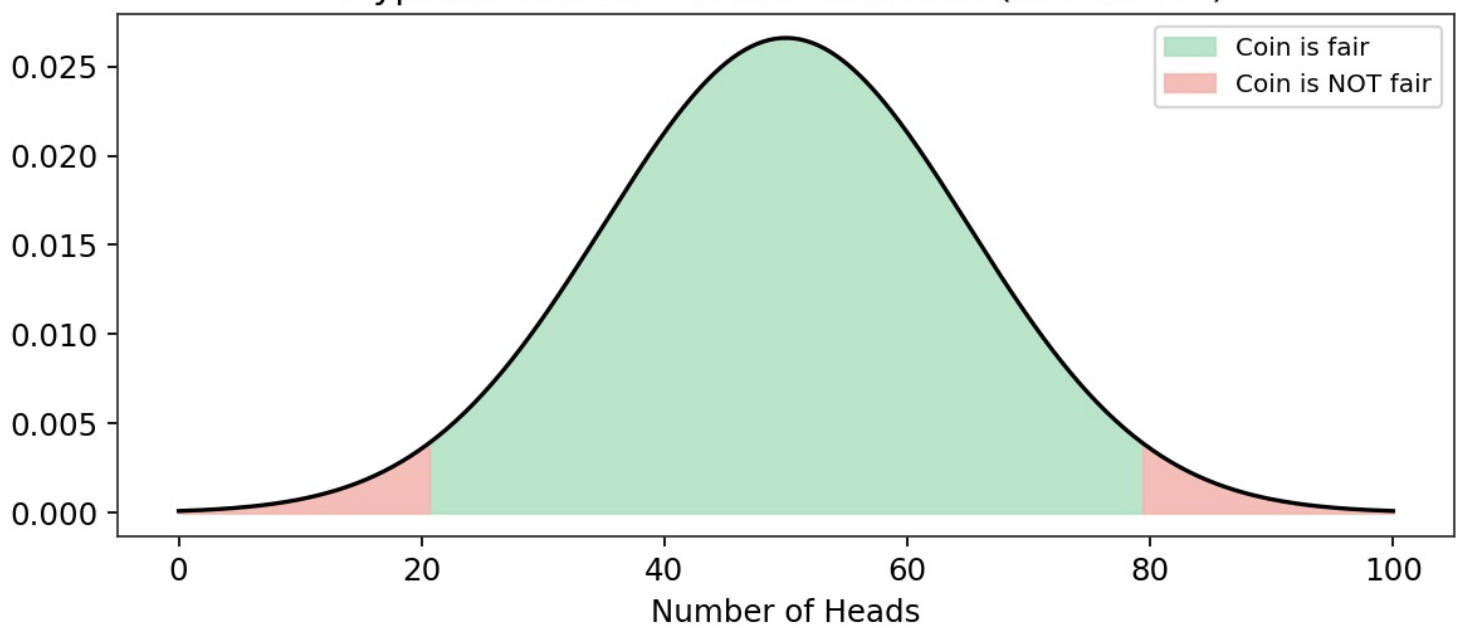


Fig 2.1 — The green "acceptance" region covers 95% of outcomes consistent with a fair coin. Extreme results (in the red tails) suggest the coin is not fair. Rejection area sits at 2.5% on each side ($0.025 + 0.025 = 0.05 = \alpha$).

Interpreting the Result

If out of 100 tosses we observe, say, 80 heads — that result falls in the red rejection zone, giving $p < 0.05$, so we **Reject H_0** and conclude the coin is likely unfair.

If we observe 52 heads — that's well within the green zone, giving $p > 0.05$, so we **Fail to Reject H_0** (not enough evidence the coin is unfair).

* Extra Example — A/B Testing a Website Button

A company tests whether a new "Buy Now" button color increases purchases. H_0 : no difference in conversion rate. After running the experiment, they get $p = 0.03$. Since $0.03 < 0.05$ (their chosen α), they **reject H_0** and conclude the new button color genuinely improves conversions — this exact logic (p-value vs α) is used millions of times a day in tech company experimentation platforms.

💡 **Common misconception:** A small p-value does NOT mean "the alternate hypothesis is true with high probability." It only means the observed data would be unlikely *if H_0 were true*. Also, a p-value is sensitive to sample size is a subtlety worth remembering, but for exam purposes the decision rule ($p < \alpha \rightarrow$ reject H_0) is what matters most.

3. Z-test

Z-test is a hypothesis test used when: **(i) the population standard deviation (σ) is known**, and **(ii) sample size $n \geq 30$** .

Family of hypothesis tests, based on the type of data:

Test	Used to test	Reference table
Z-test	Averages/means (σ known, $n \geq 30$)	Z-table $\rightarrow$ Z-score & p-value
t-test	Averages/means (σ unknown)	t-table
Chi-Square	Categorical data	Chi-Square table
ANOVA	Variance across 3+ groups	F-table

Z-score for a sample mean: $Z_d = (\bar{x} - \mu) / (\sigma/\sqrt{n})$

Worked Example — City Heights (Two-Tailed Test)

The average height of all residents in a city is $\mu = 168$ cm, $\sigma = 3.9$. A doctor believes the mean is different. He measures 36 individuals and finds $\bar{x} = 169.5$ cm.

(a) $H_0: \mu = 168$ cm $H_1: \mu \neq 168$ cm (two-tailed test) $CI = 95\%$, $\alpha = 1 - 0.95 = 0.05$

(b) Calculate Z:

$Z_d = (169.5 - 168) / (3.9/\sqrt{36}) = 1.5 / 0.65 = \mathbf{2.31}$

Two-Tailed Test — Rejection Regions at $\alpha = 0.05$

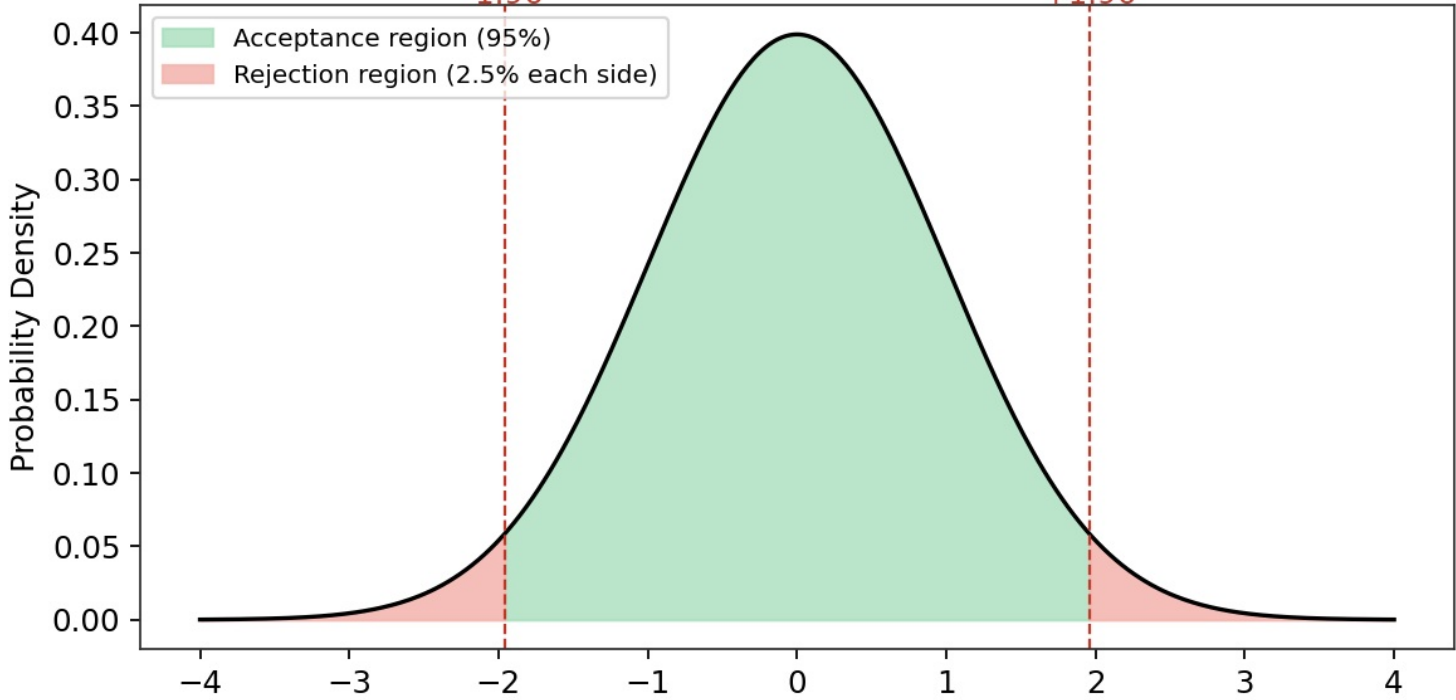


Fig 3.1 — Two-tailed rejection regions at $\alpha=0.05$: critical Z-values are ± 1.96 .

Decision & Conclusion

Since $|Z| = 2.31 > 1.96 \rightarrow$ falls in the rejection area $\rightarrow$ **Reject the Null Hypothesis**.

Equivalently: $p\text{-value} = 0.0104 + 0.0104 = 0.0208$, and since $0.0208 < 0.05 \rightarrow$ **Reject H_0** .

Final Conclusion: The average height of this city seems to be increasing compared to 168 cm, based on the sample.

Worked Example — One-Tailed Test (Light Bulbs)

A factory manufactures bulbs with average warranty $\mu = 5$ years, $\sigma = 0.50$. A worker believes bulbs will malfunction in **less than** 5 years. Sample of 40 bulbs gives $\bar{x} = 4.8$ years. Test at $\alpha = 0.02$.

$H_0: \mu = 5$ $H_1: \mu < 5$ (one-tailed / left-tailed test)

$Z_d = (4.8 - 5) / (0.50/\sqrt{40}) = -0.2/0.079 = \mathbf{-2.53}$

One-Tailed (Left) Test — Rejection Region at $\alpha = 0.02$

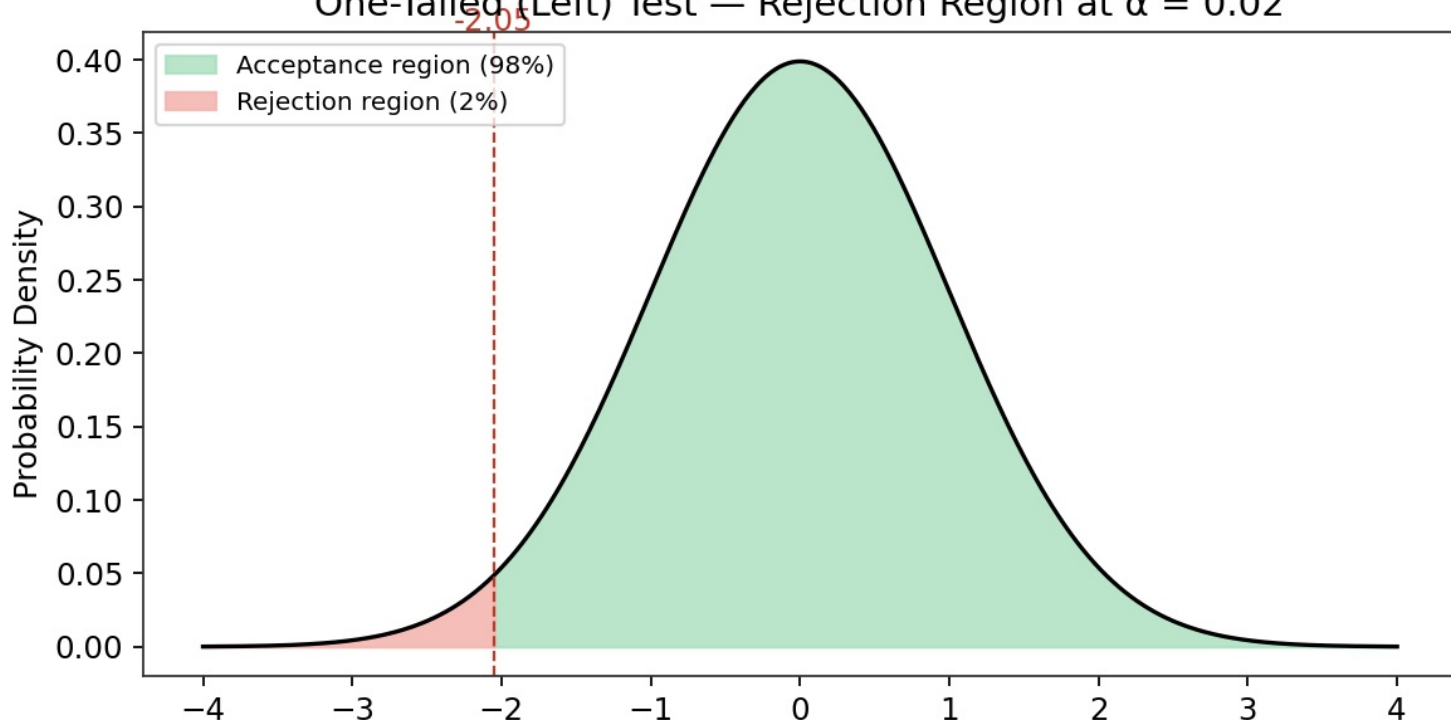


Fig 3.2 — One-tailed (left) rejection region at $\alpha=0.02$: critical Z-value is -2.05 .

Decision

Area under curve at $Z = -2.53$ gives p-value = 0.0570. Compare with $\alpha = 0.02$: since $0.0570 < 0.02$ is **FALSE** (0.0570 is actually greater), we **Accept (fail to reject) the Null Hypothesis** — not enough evidence the warranty should be revised.

* Extra Example — Factory Bolt Diameter

A machine is calibrated to produce bolts with diameter $\mu = 10\text{mm}$, $\sigma = 0.2\text{mm}$ (known from years of historical production data — hence Z-test applies). QA takes a sample of $n=50$ bolts and gets $\bar{x} = 10.08\text{mm}$. Test at $\alpha=0.05$, two-tailed.

$Z = (10.08 - 10) / (0.2 / \sqrt{50}) = 0.08 / 0.0283 = 2.83$. Since $2.83 > 1.96 \rightarrow$ **Reject H_0** — the machine may be miscalibrated and drifting off-spec.

4. Student's t-Distribution

The Problem: In Z-statistics, any analysis using Z-score requires that σ (population standard deviation) is **already known**. But what do we do when we **don't** know the population standard deviation? → Enter **Student's t-distribution**.

Z-score: $Z = (\bar{x} - \mu) / (\sigma/\sqrt{n})$ → **needs population σ , uses Z-table**

t-statistic: $t = (\bar{x} - \mu) / (S/\sqrt{n})$ → **uses sample std. dev S , uses t-table (t-test)**

4.1 Degrees of Freedom (dof)

$$\text{dof} = n - 1$$

Intuition: With 3 people, once you fix any 2 of their values (given the mean), the 3rd value is automatically determined — it has no "freedom" left. So only $n-1$ values are truly "free" to vary. Example: $n = 3 \rightarrow \text{dof} = 3 - 1 = 2$.

Student's t-Distribution vs Standard Normal

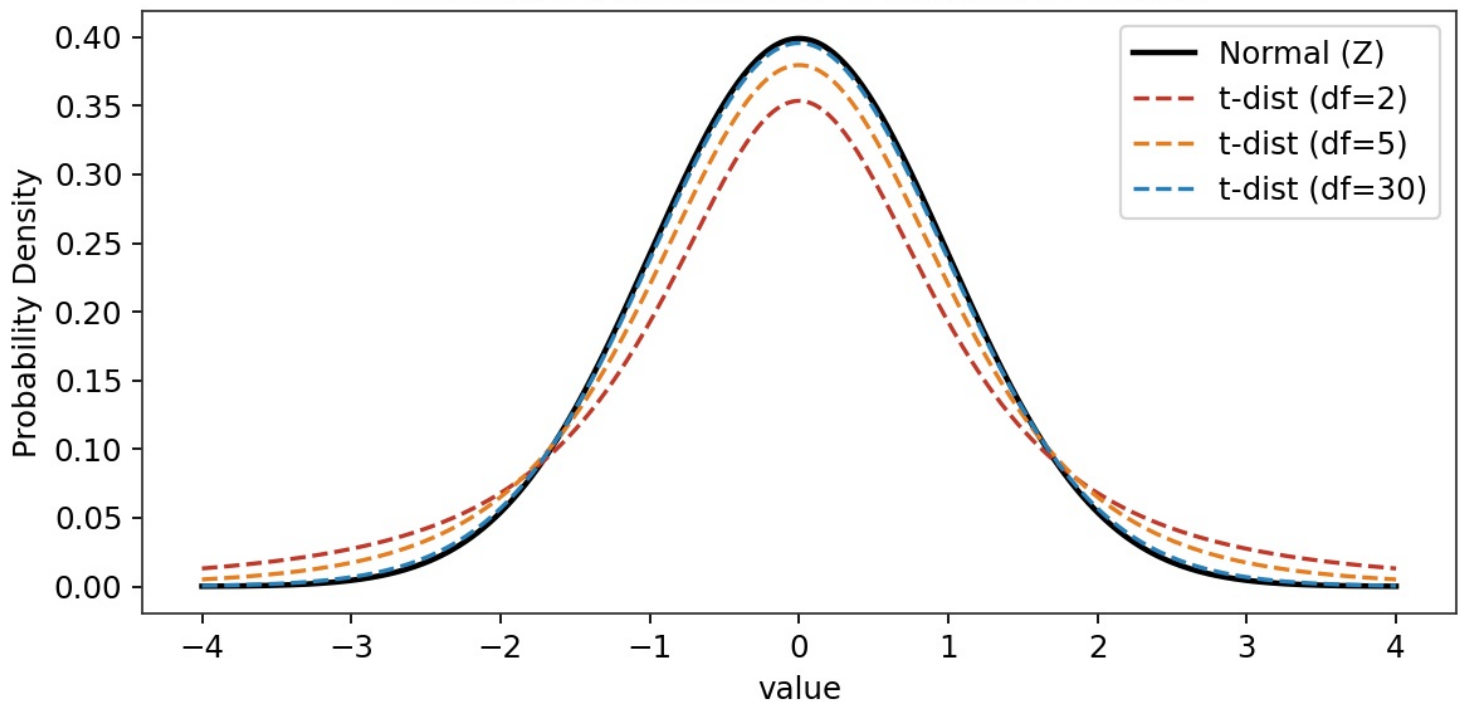


Fig 4.1 — The t-distribution has fatter tails than the Normal (Z) distribution, especially for small degrees of freedom. As df grows large ($\text{df} \geq 30$), the t-distribution converges to the Normal distribution.

💡 **Why fatter tails?** With small samples, our estimate of the standard deviation (S) is less reliable, so the t-distribution builds in extra uncertainty — making extreme values slightly more likely than the Normal distribution predicts.

✳ Extra Example — When dof Matters

Suppose you test the average commute time for 5 employees ($n=5$, $\text{dof}=4$) vs. 100 employees ($n=100$, $\text{dof}=99$). With only 5 data points, the t-distribution used is wide and flat (much uncertainty). With 100 data points, the t-distribution is nearly identical to the Normal curve — this is exactly why t-tests are essential for small-sample research (like early-stage clinical trials) but Z-tests/large-sample t-tests work fine for big datasets.

5. T-stats & One-Sample t-test

T-stats ÷ (drives) the **One-Sample t-test** — used to test a sample mean against a known/claimed population mean when σ is unknown.

$$t = (\bar{x} - \mu) / (S/\sqrt{n})$$

5.1 Worked Example — Does Medication Affect IQ?

Population average IQ $\mu = 100$. Researchers test a new medication on a sample of $n=30$ participants. Sample mean $\bar{x} = 140$, sample std. dev $S = 20$. Test at 95% CI, $\alpha = 0.05$.

1. **Null Hypothesis H_0 :** $\mu = 100$ **Alternate H_1 :** $\mu \neq 100$ (2-tailed test)
2. **$\alpha = 0.05$**
3. **Degrees of Freedom:** $\text{dof} = n - 1 = 30 - 1 = 29$
4. **Decision Rule:** If $t < -2.045$ or $t > +2.045 \rightarrow \text{Reject } H_0$

One-Sample t-Test — Decision Boundary (df = 29)

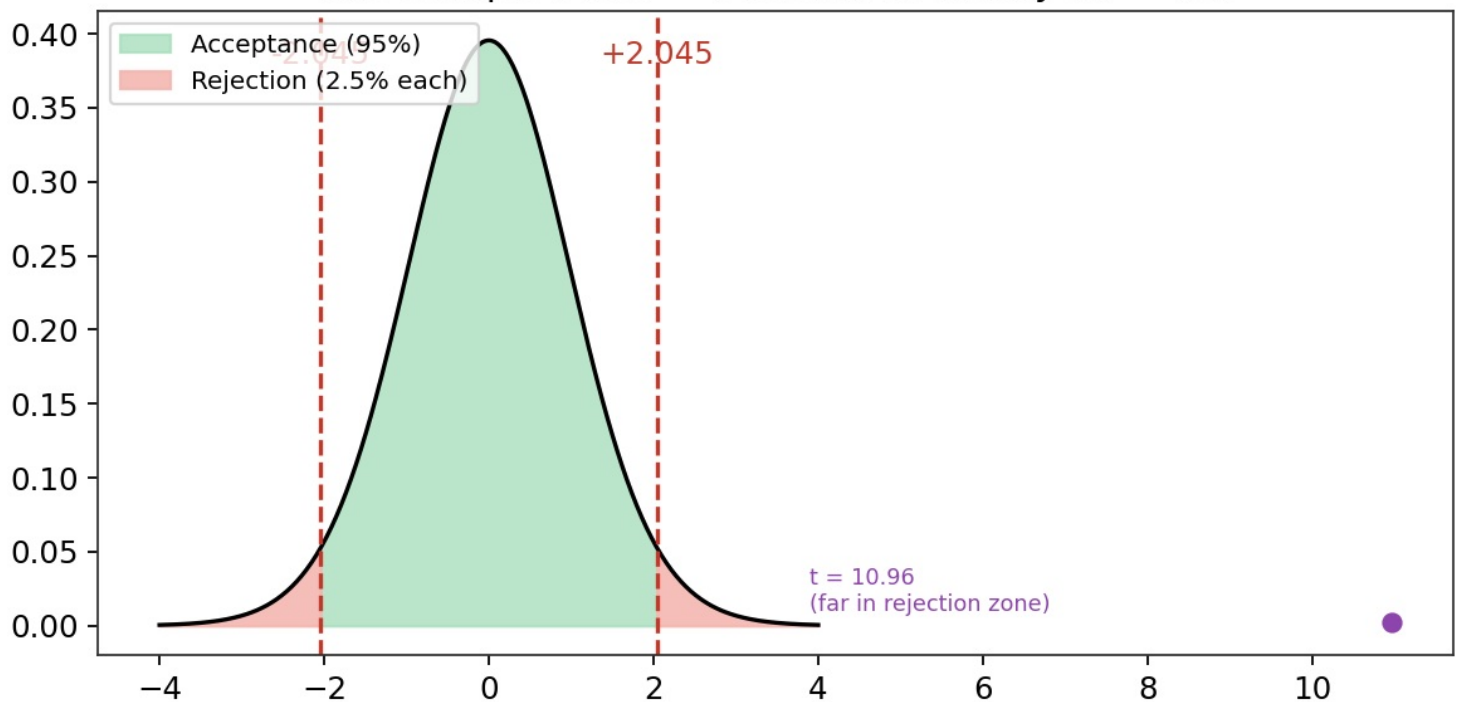


Fig 5.1 — One-sample t-test decision boundary at $\text{df}=29$: critical values ± 2.045 . Our computed $t=10.96$ lands far into the rejection zone.

Calculate Test Statistic

$$t = (140 - 100) / (20/\sqrt{30}) = 40 / 3.65 = \mathbf{10.96}$$

Since $t = 10.96 > 2.045 \rightarrow \text{Reject the Null Hypothesis.}$

Conclusion: The medication used has affected intelligence — specifically, it has increased intelligence.

* Extra Example — Average Delivery Time

An online store claims average delivery time is $\mu = 3$ days. A customer watchdog group samples $n=25$ recent deliveries and finds $\bar{x} = 3.8$ days, $S = 1.2$ days. Test at $\alpha=0.05$ (2-tailed, $\text{dof}=24$).

$t = (3.8-3)/(1.2/\sqrt{25}) = 0.8/0.24 = \mathbf{3.33}$. Critical value for $\text{df}=24$ at $\alpha=0.05$ (two-tail) $\approx \pm 2.064$. Since $3.33 > 2.064 \rightarrow \text{Reject } H_0$ — actual delivery times are significantly longer than claimed.

🔗 **T-stats "J-test" note:** The one-sample t-test is sometimes just called the "t-test" — always check whether you're comparing a sample mean to a fixed value (one-sample) or comparing two sample means to each other (two-sample / paired t-test), since the formulas differ slightly.

6. Z-test vs T-test — When to Use Which

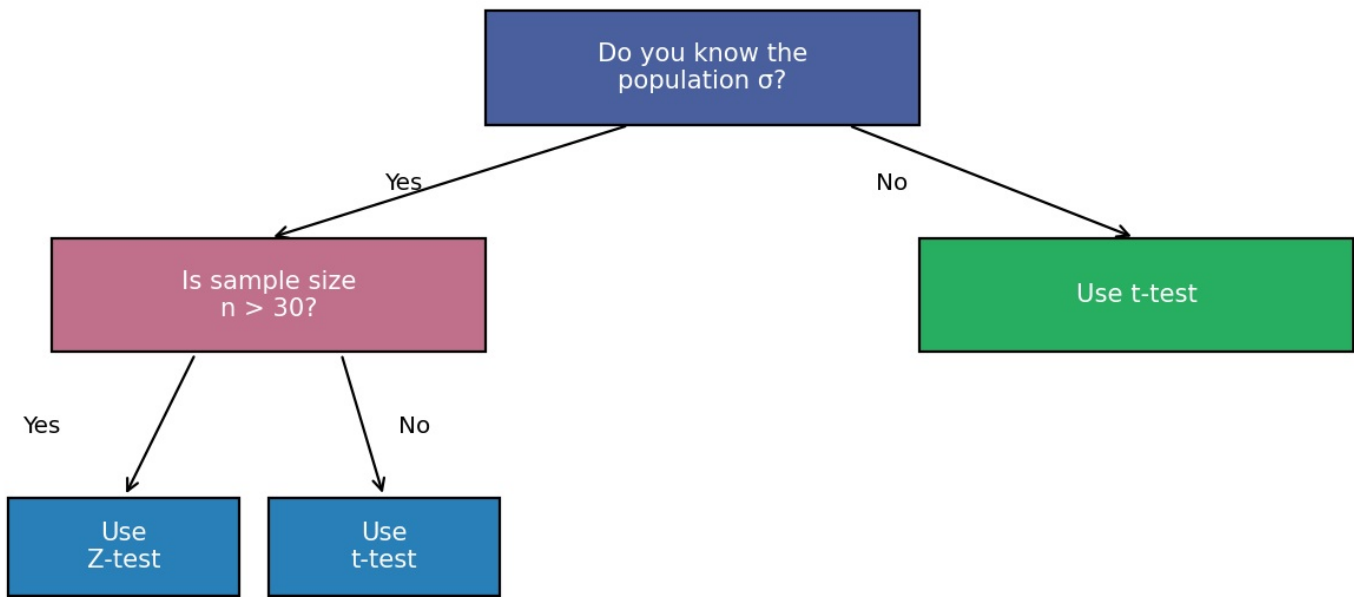


Fig 6.1 — Decision flowchart: population σ known → check sample size; population σ unknown → always use t-test.

Criterion	Z-test	t-test
Population σ known?	Yes	No
Sample size	$n \geq 30$ (large)	Any size, but especially $n < 30$ (small)
Distribution used	Standard Normal (Z)	Student's t (fatter tails)
Formula	$Z = (\bar{x} - \mu) / (\sigma / \sqrt{n})$	$t = (\bar{x} - \mu) / (S / \sqrt{n})$
Reference table	Z-table	t-table (uses dof = $n - 1$)

💡 **Simple decision rule:** Do you know the population standard deviation (σ)? → **Yes:** Is the sample size above 30? → Yes: use Z-test | No: use t-test. → **No** (σ unknown): always use t-test, regardless of sample size.

✳ Extra Example — Choosing the Right Test

- **Scenario A:** A national census bureau knows the exact population σ for household income (from decades of full census data) and samples $n=500$ households → **Z-test** (σ known, large n).
- **Scenario B:** A startup tests a new feature on only $n=12$ beta users and has no prior knowledge of the true variability → **t-test** (σ unknown, small n).
- **Scenario C:** A university doesn't know the true population σ of exam scores but samples $n=200$ students → **t-test** is technically correct (σ unknown), though with such a large sample, the t-distribution is nearly identical to Z anyway.

7. Type I and Type II Errors

Reality can be: Null Hypothesis is True, OR Null Hypothesis is False.
Our Decision can be: Null Hypothesis is True, OR Null Hypothesis is False.
Combining these two gives **4 possible outcomes**:

Outcome	Our Decision	Reality	Result
Outcome 1	Reject H_0	H_0 is actually False	✔ Good (Correct)
Outcome 2	Reject H_0	H_0 is actually True	✗ Type I Error (α)
Outcome 3	Retain (fail to reject) H_0	H_0 is actually False	✗ Type II Error (β)
Outcome 4	Retain H_0	H_0 is actually True	✔ Good (Correct)

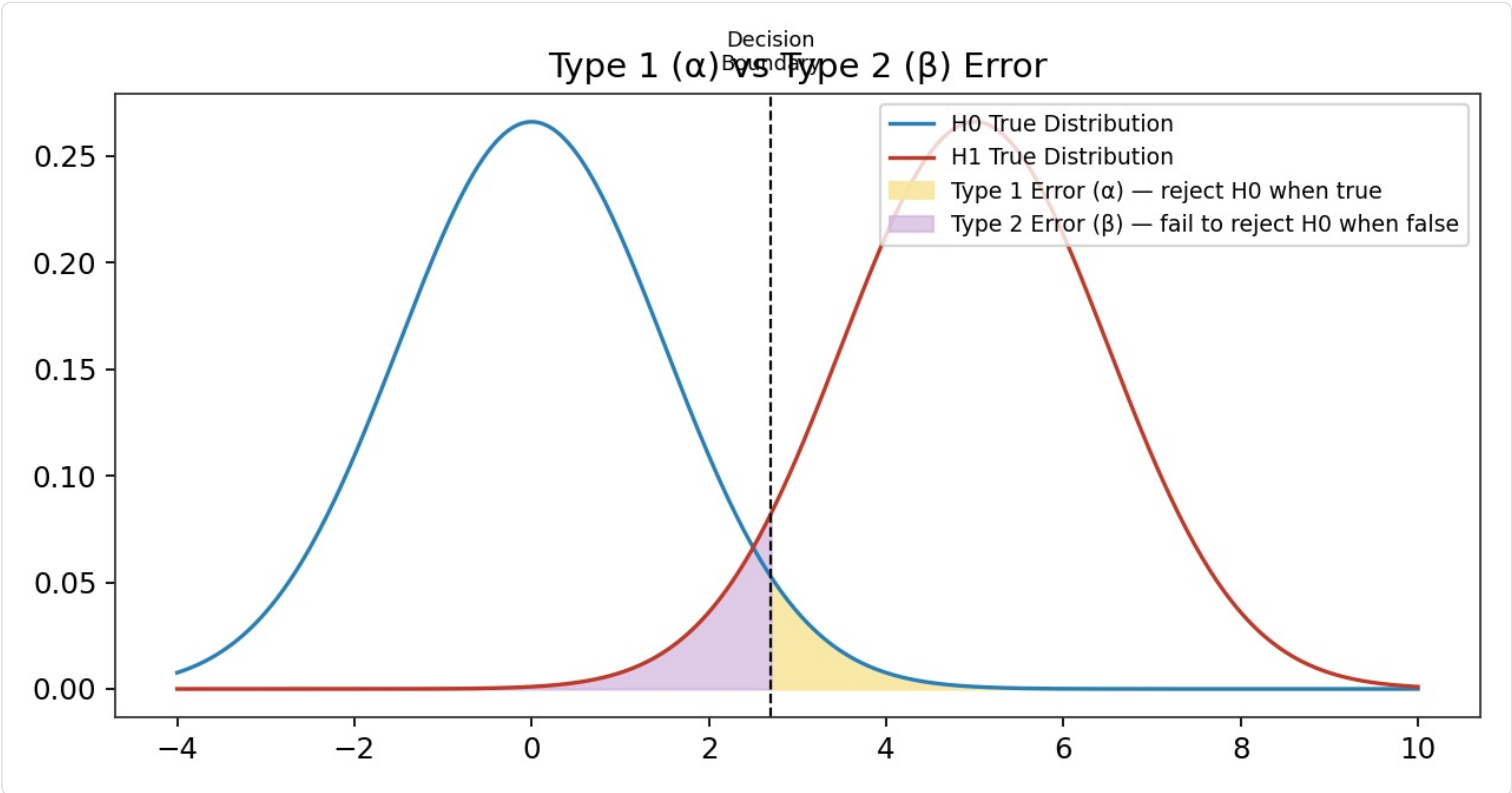


Fig 7.1 — Type I error (α , yellow) = rejecting H_0 when it's actually true. Type II error (β , purple) = failing to reject H_0 when it's actually false. The two distributions represent "reality under H_0 " and "reality under H_1 ."

Memory trick — courtroom analogy:

- **Type I Error** = convicting an innocent person (false alarm) — we reject H_0 ("not guilty") when they really were innocent.
- **Type II Error** = letting a guilty person go free (missed detection) — we fail to reject H_0 when the person was actually guilty.

Worked Example — Medical Testing

H_0 : "Patient does not have the disease."

- **Type I Error:** Test says patient HAS the disease, but they actually don't (false positive) → causes unnecessary stress/treatment.
- **Type II Error:** Test says patient does NOT have the disease, but they actually do (false negative) → dangerous, disease goes untreated.

✳ Extra Example — Spam Filters

H_0 : "This email is not spam."

- **Type I Error:** A genuine important email gets marked as spam (false positive) — annoying, you might miss a job offer!
- **Type II Error:** An actual spam/phishing email lands in your inbox (false negative) — potentially dangerous.

Notice the tradeoff: making a spam filter super aggressive reduces Type II errors but increases Type I errors, and vice versa. This exact tradeoff (α vs β) appears throughout statistics, medicine, and machine learning (precision vs recall).

🔑 **Key relationship:** Lowering α (being more strict about rejecting H_0) typically increases β (Type II error), and vice versa — there's an inherent tradeoff. The only way to reduce both simultaneously is to increase the sample size.

8. Bayes' Theorem

Bayesian statistics is an approach to data analysis and parameter estimation based on **Bayes' Theorem** — a rule for updating probabilities when new information (evidence) becomes available.

8.1 Independent vs Dependent Events

Type	Meaning	Example
Independent Events	One event doesn't affect the other's probability	Rolling a die {1,...,6}: $\Pr(1)=1/6$ always. Tossing a coin: $\Pr(H)=0.5$ always
Dependent Events	One event's outcome changes the other's probability	Drawing colored balls without replacement

8.2 Conditional Probability — The Box Example

A box has balls: some Red, some other colors. Among Red balls, some are also Yellow (two-color balls). Given $\Pr(\text{Red}) = 2/5$ and $\Pr(\text{Yellow} \mid \text{Red}) = 3/4$:

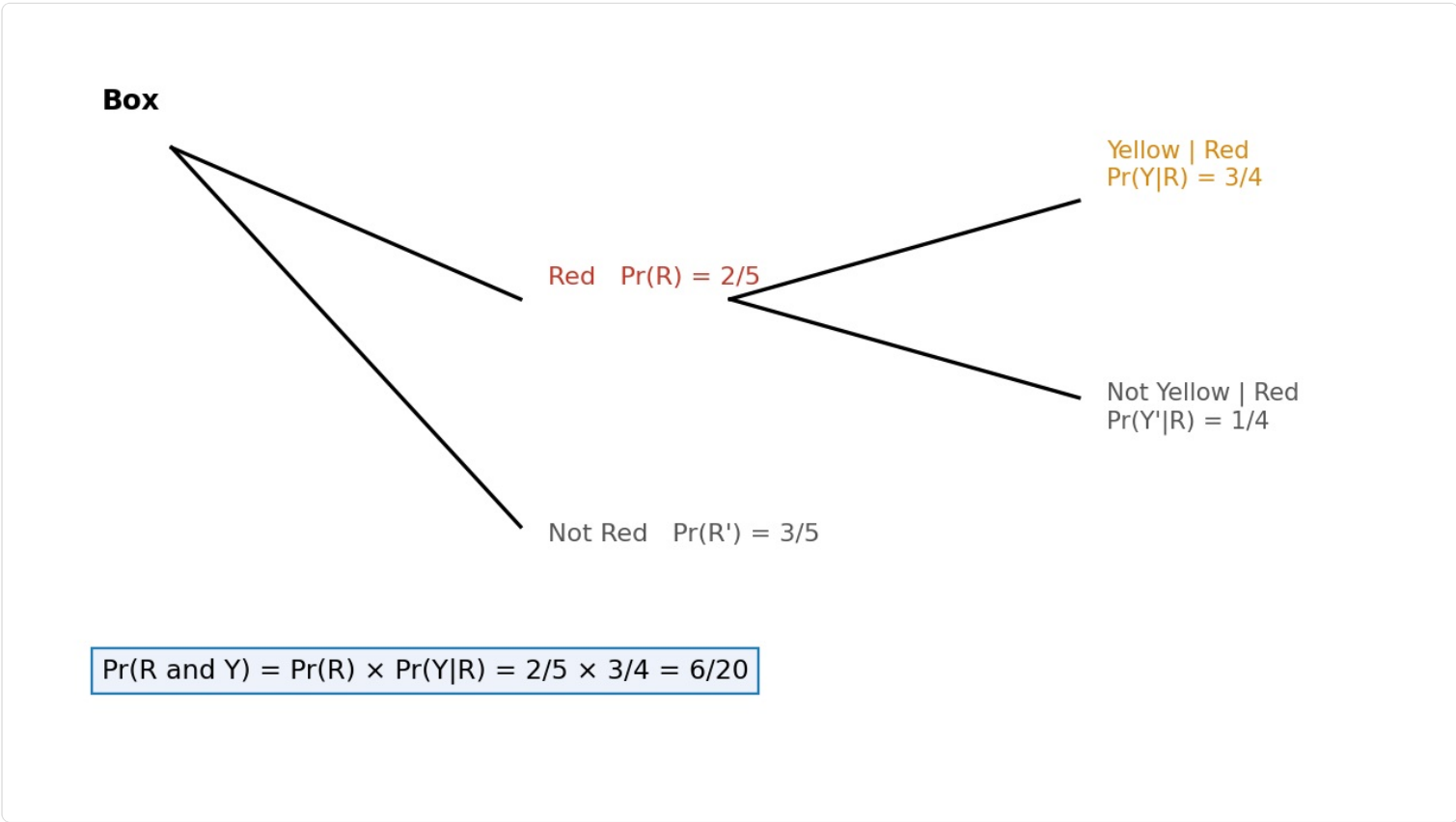


Fig 8.1 — Probability tree: first branch by color (Red/Not Red), second branch by a second attribute (Yellow/Not Yellow) given Red.

$$\begin{aligned} \Pr(R \text{ and } Y) &= \Pr(R) \times \Pr(Y \mid R) \quad (\text{Conditional Probability}) \\ &= 2/5 \times 3/4 = 6/20 \end{aligned}$$

8.3 Deriving Bayes' Theorem

$$\begin{aligned} \Pr(A \text{ and } B) &= \Pr(B \text{ and } A) \\ \Pr(A) \cdot \Pr(B|A) &= \Pr(B) \cdot \Pr(A|B) \end{aligned}$$

$$\Rightarrow \Pr(B|A) = [\Pr(B) \times \Pr(A|B)] / \Pr(A)$$

Bayes' Theorem: $\Pr(A|B) = [\Pr(A) \times \Pr(B|A)] / \Pr(B)$

where A, B are events; $\Pr(A|B)$ = probability of A given B is true; $\Pr(A)$, $\Pr(B)$ = independent (marginal) probabilities of A and B.

Worked Example — Rare Disease Testing

1% of a population has a disease: $\Pr(\text{Disease}) = 0.01$. A test is 95% accurate: $\Pr(\text{Positive} \mid \text{Disease}) = 0.95$, and has a 5% false positive rate: $\Pr(\text{Positive} \mid \text{No Disease}) = 0.05$.
What is $\Pr(\text{Disease} \mid \text{Positive})$?

$$\begin{aligned} \Pr(\text{Positive}) &= \Pr(\text{Positive}|\text{Disease}) \times \Pr(\text{Disease}) + \Pr(\text{Positive}|\text{No Disease}) \times \Pr(\text{No Disease}) \\ &= 0.95 \times 0.01 + 0.05 \times 0.99 = 0.0095 + 0.0495 = 0.059 \end{aligned}$$

$\Pr(\text{Disease} \mid \text{Positive}) = (0.95 \times 0.01) / 0.059 = 0.0095/0.059 \approx \mathbf{16.1\%}$

Surprising insight: Even with a positive test result, there's only a ~16% chance the person actually has the disease! This is because the disease is so rare (1%) that false positives from the healthy 99% majority outnumber true positives — a classic and important Bayesian result.

8.4 Bayes' Theorem in Data Science (Feature-Based Prediction)

Consider a House Price dataset with features: Size (x_1), No. of Rooms (x_2), Location (x_3) — all independent — and target output Price (y), which is dependent on all of them:

$$\Pr(y \mid x_1, x_2, x_3) = [\Pr(y) \times \Pr(x_1, x_2, x_3 \mid y)] / \Pr(x_1, x_2, x_3)$$

This is exactly the mathematical foundation of the **Naive Bayes Classifier**, a popular ML algorithm — it uses Bayes' theorem (with a "naive" assumption that features are conditionally independent) to predict a target class from input features.

✳ Extra Example — Naive Bayes for Spam Detection

Given the word "FREE" appears in an email, we want $\Pr(\text{Spam} \mid \text{"FREE" appears})$.

$\Pr(\text{Spam}) = 0.4$ (40% of all emails are spam), $\Pr(\text{"FREE"} \mid \text{Spam}) = 0.7$, $\Pr(\text{"FREE"} \mid \text{Not Spam}) = 0.05$.

$\Pr(\text{"FREE"}) = 0.7 \times 0.4 + 0.05 \times 0.6 = 0.28 + 0.03 = 0.31$

$\Pr(\text{Spam} \mid \text{"FREE"}) = (0.7 \times 0.4) / 0.31 = 0.28 / 0.31 \approx \mathbf{90.3\%}$ — a very strong signal that emails containing "FREE" are spam. This exact calculation, repeated across thousands of words, is how early spam filters (and Naive Bayes classifiers in general) work.

9. Master Summary Table

Concept	Key Formula	When Used
Z-score / Z-test	$Z = (\bar{x} - \mu) / (\sigma / \sqrt{n})$	σ known, $n \geq 30$
t-statistic / t-test	$t = (\bar{x} - \mu) / (S / \sqrt{n})$	σ unknown, any n ($dof = n - 1$)
P-value	Compare to α	Decide reject/fail-to-reject H_0
Type I Error (α)	$\Pr(\text{Reject } H_0 \mid H_0 \text{ true})$	False positive / false alarm
Type II Error (β)	$\Pr(\text{Fail to reject } H_0 \mid H_0 \text{ false})$	False negative / missed detection
Bayes' Theorem	$\Pr(A B) = \Pr(A)\Pr(B A) / \Pr(B)$	Updating probability given new evidence

 **Quick Recall Checklist**

- H_0 = starting assumption (innocent until proven guilty); H_1 = opposite of H_0
- p-value < α (significance level, commonly 0.05) → Reject H_0
- Z-test needs known population σ AND $n \geq 30$; t-test used when σ is unknown
- t-distribution $\approx$ Normal distribution as degrees of freedom ($dof = n - 1$) grow large
- Two-tailed test → H_1 uses " $\neq$ "; One-tailed test → H_1 uses "<" or ">"
- Type I Error (α) = false positive (reject true H_0); Type II Error (β) = false negative (retain false H_0)
- Bayes' Theorem: $\Pr(A|B) = \Pr(A) \cdot \Pr(B|A) / \Pr(B)$ — updates belief given new evidence
- Naïve Bayes Classifier = ML algorithm built directly on Bayes' Theorem